HHT–BAO Locking Analysis Report

Version: 1V4

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Executive Summary

This update builds on versions 1V0–1V3 of the HHT–BAO locking analysis.

The most recent analyses extend the Noether soft-breaking grid, introduce harmonic ratio tests, and explores redshift-window dependencies.

Key results show that IMF2 remains the primary carrier of BAO locking at low Noether strengths, but decoheres at higher values.

IMF3, in contrast, demonstrates conditional behavior: independent in most contexts, but harmonic (1:2) under specific λ-driven scenarios.

Results Update

1. Extended Noether Grid (s = 0–12)

- IMF2: Stable locking at s = 0–4; decoheres at s = 6 and 10; partial recovery at s = 8.
- IMF3: Weak baseline signal; flips correlation sign at s = 6–8; acts independent at higher strengths.

Reference files:
- noether_grid_IMF2_IMF3_vs_strength.csv
- noether_grid_summary_direction_scores.csv

2. Harmonic Ratio Tests (IMF3 / IMF2)

- λ_on: Mean ratio ≈ 0.5 → harmonic locking.
- λ_grid: Mean ratio ≈ 1.0 → independent oscillators.
- Baseline: Scatter, wide variance around −0.5.
- Conserve_time: ~0.06, decoupled.

Reference files:
- harmonic_test_harmonic_test.csv
- harmonic_test_ratio_summary.png

3. Redshift Window Dependencies

- Full window ($z = 0.02–1.9$): IMF2 modest lock ($p \approx 0.06$), IMF3 insignificant.
- Restricted ($z = 0.05–1.5$): IMF2 weak ($p \approx 0.91$), IMF3 non-significant.
- Window ($z = 0.10–1.2$): signals nearly vanish.

Interpretation: Locking strongest only when full redshift range is retained.

Reference file:
- hht_sn_FRPBH_windows_IMF2_IMF3_by_window.csv

4. DM vs DH Fits

- IMF2: Consistent DM/DH correlations at low strengths.
- IMF3: Opposing DM vs DH tendencies.
- Δr highlights systematic IMF3 tension.

Reference files:
- bao_compare_dm_dh_freq_scale_fits.csv
- bao_compare_dm_dh_summary.md
- noether_grid_dm_dh_soft0_freq_scale_fits.csv
- noether_grid_dm_dh_soft10_freq_scale_fits.csv

Interpretation

- IMF2 is the most stable BAO-lock carrier but fragile under Noether perturbations or redshift cuts.

- IMF3 is conditionally harmonic but often independent or oppositional.

- Harmonic ratio results reinforce distinct dynamical regimes driven by $\Lambda(z)$ or coupling assumptions.

Pros and Cons of HHT Approach (Update)

Pros:
- Sensitive to quasi-periodic microstructures in SN residuals.
- Detects conditional harmonic relations (IMF3/IMF2 ≈ 0.5).
- Flexible across redshift windows and parameter sweeps.

Cons:
- Locking signals fragile under data cuts.
- Dependent on smoothing and permutation count.
- IMF3 often noisy, challenging interpretation.

Next Steps

1. Extend harmonic ratio analysis using raw instantaneous frequencies.
2. Introduce Planck priors and BAO covariance into HHT runs.
3. Apply wavelet decomposition as cross-check.
4. Rank FR, PBH, and ΛCDM models by AIC/BIC with HHT signals.
5. Investigate IMF3's oppositional DM vs DH behavior.

Reference Dataset: Final Curated Drop

Curated results: /Users/boyde/Desktop/HHT_BAO_FINAL_DROP_CLEAN/

Key files:
- hht_bao_lock_real_baseline_s9_bao_lock_heatmap_table_with_p.csv
- noether_grid_IMF2_IMF3_vs_strength.csv
- noether_grid_summary_direction_scores.csv
- bao_compare_dm_dh_freq_scale_fits.csv
- bao_compare_dm_dh_summary.md
- harmonic_test_harmonic_test.csv
- harmonic_test_ratio_summary.png
- hht_sn_FRPBH_windows_IMF2_IMF3_by_window.csv